

XUA Gold Price Dataset Analysis

Presented By:

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Course : Data Science And Machine Learning

INTRODUCTION

In modern financial markets, gold is traded as a commodity, and its price is represented by **XAU**, which stands for one troy ounce of gold. The XAU Gold Price Dataset contains historical records of gold prices over a specific period, providing valuable insights into market trends and price movements.

This dataset is an example of **time-series data**, where observations are recorded at regular intervals (daily, weekly, or monthly). It helps in understanding how gold prices fluctuate over time due to various economic factors such as inflation, interest rates, geopolitical tensions, currency strength (especially the US dollar), and global demand and supply

Gold Market Intelligence Dashboard (2015-2025)

1.76K

avg_close

4.53K

max_close

1.06K

min_close

40.33

Volatility

4.32K

Current Price

Year

2015

2021

2016

2022

2017

2023

2018

2024

2019

2025

2020

Yearly Gold Price Volatility



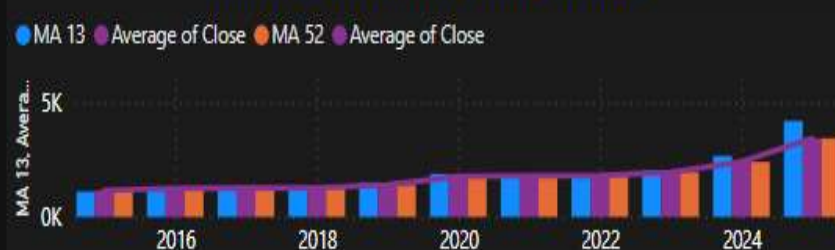
Monthly Avg of Gold Price



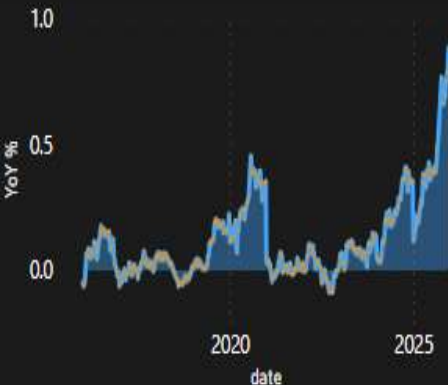
Weekly Gold Price Change



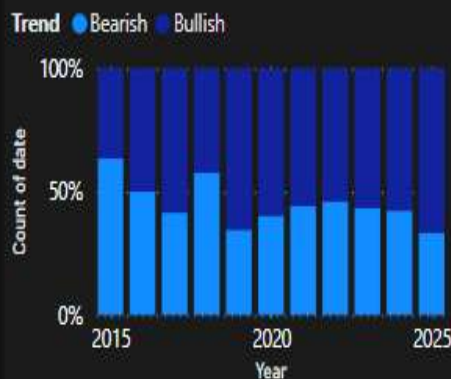
Gold Price Trend With Moving Average



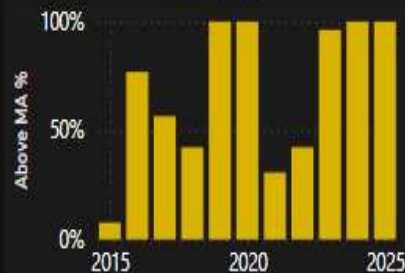
Year-to-Year Growth



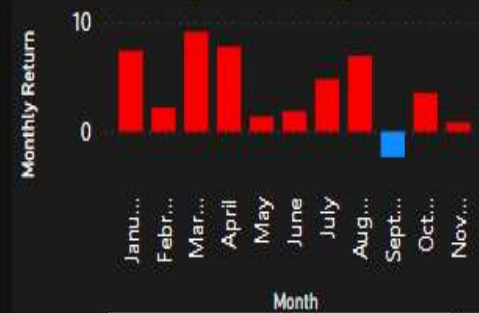
Bullish vs Bearish Weeks



Gold Price Stayed Above Moving Average

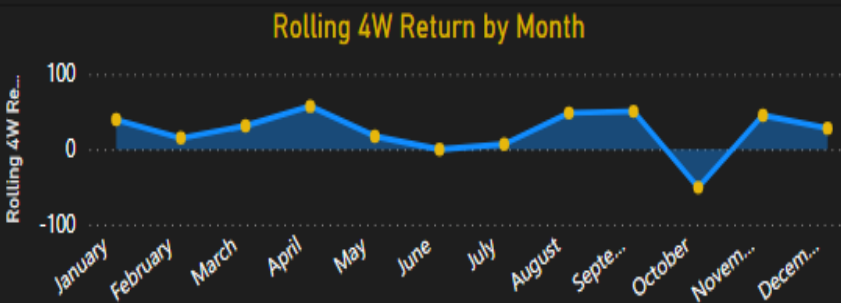
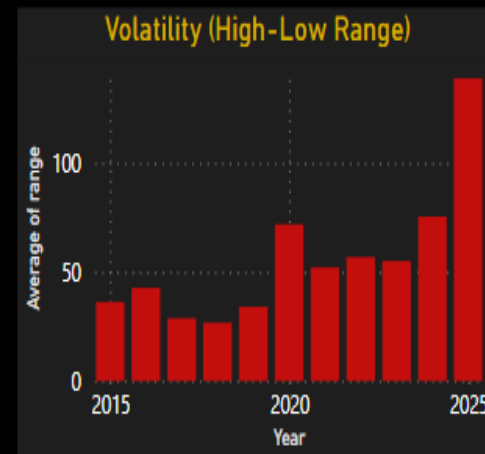
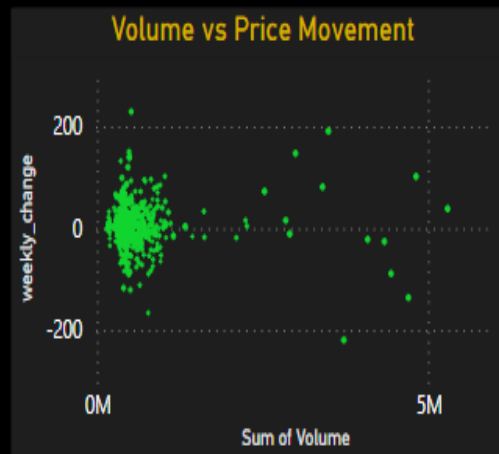
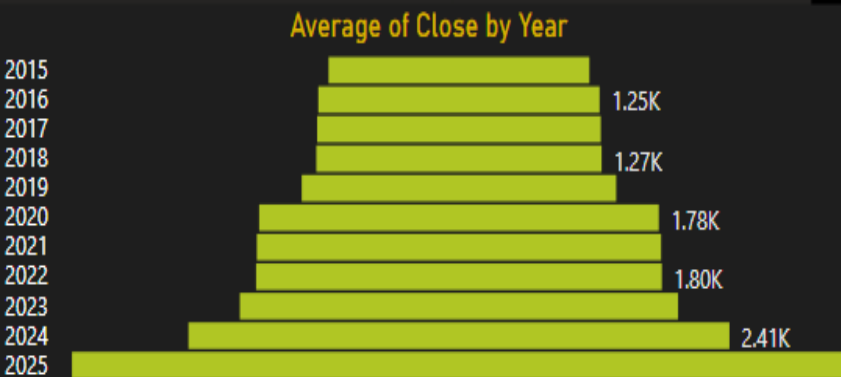


Monthly Return Analysis



Trend, Volatility and Momentum Analysis

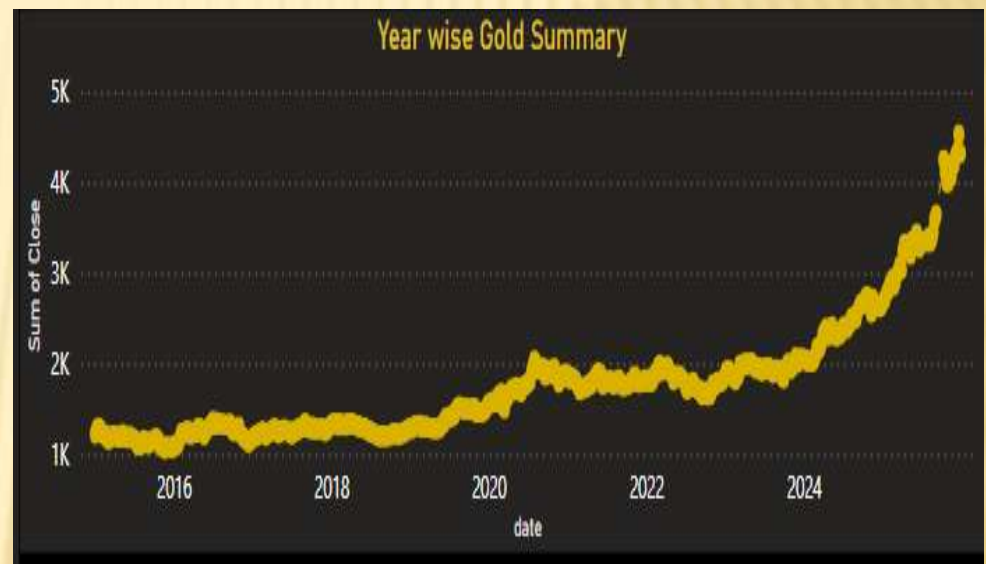
Year	Average of Close	MA 13	MA 52_2	Year	Sum of Close	Sum of Open
2015	1,159.57	1,100.93	1,159.57	2015	60,297.81	60,456.40
2016	1,250.60	1,209.83	1,250.60	2016	65,031.45	64,915.50
2017	1,261.28	1,281.97	1,262.98	2017	66,847.58	66,674.07
2018	1,268.68	1,237.87	1,268.68	2018	65,971.48	66,023.30
2019	1,398.14	1,486.36	1,398.14	2019	72,703.06	72,457.80
2020	1,776.79	1,879.92	1,776.79	2020	92,392.88	92,133.52
2021	1,797.48	1,801.49	1,797.48	2021	93,468.89	93,577.78
2022	1,804.12	1,739.14	1,804.12	2022	93,814.29	93,859.75
2023	1,948.37	2,006.82	1,949.96	2023	1,03,263.74	1,03,032.10
2024	2,405.27	2,664.55	2,405.27	2024	1,25,074.20	1,24,480.44
2025	3,441.11	4,201.26	3,378.97	2025	1,65,173.13	1,64,008.40
Total	1,761.47	4,201.26	3,378.97	Total	10,04,038.51	10,01,619.06



Q1. HOW HAS THE OVERALL GOLD PRICE TRENDED FROM 2015 TO 2025?

Insight:

Over the period from 2015 to 2025, gold prices have demonstrated a clear long-term upward trend, reflecting its role as a reliable store of value. In the early years (2015–2017), prices remained relatively stable with moderate growth. However, from 2019 onwards, there was a significant increase in prices, particularly during 2020 due to global uncertainty caused by the COVID-19 pandemic. The highest closing prices were observed in the later years (2024–2025), indicating strong investor demand. The lowest prices were recorded in the initial years (2015–2016). Overall, the trend confirms that gold acts as a safe-haven asset and appreciates over time.



Q2. WHICH WEEKS SAW THE MOST DRAMATIC PRICE CHANGES?

Insight:

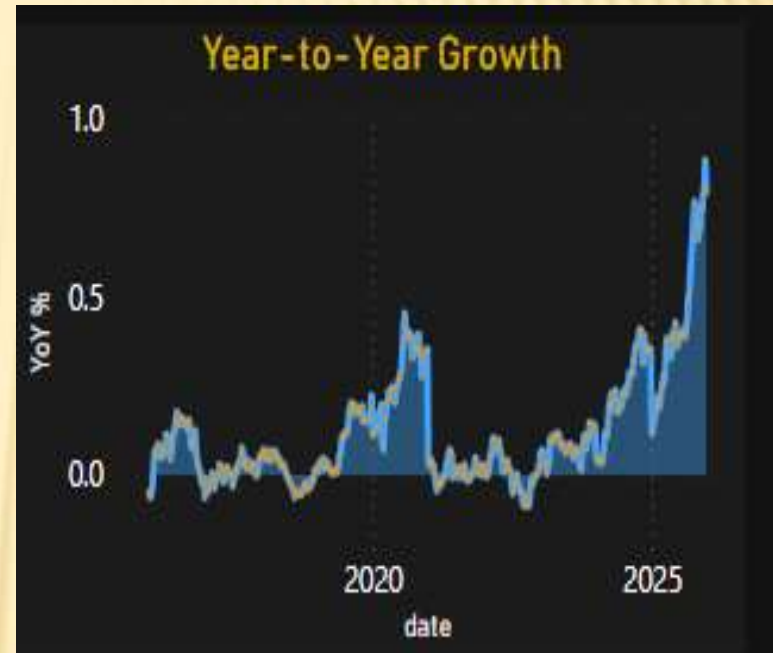
The most dramatic weekly price movements were observed during periods of global instability, particularly in 2020 and 2022. During the COVID-19 crisis in 2020, gold prices experienced sharp increases as investors shifted towards safer assets. Similarly, in 2022, inflationary pressures and interest rate changes led to sudden fluctuations in gold prices. These dramatic weekly movements highlight the sensitivity of gold prices to macroeconomic factors and global events.



Q3. WHAT IS THE YEAR-OVER-YEAR (YOY) GROWTH IN GOLD PRICES?

Insight:

The year-over-year analysis shows that gold prices generally experienced positive growth across the decade, with a few exceptions during stable economic periods. The most significant growth occurred in 2020, driven by increased demand for safe investments during economic uncertainty. In contrast, years with stable economic conditions showed slower growth or minor declines. This pattern indicates that gold performs particularly well during periods of financial instability.



Q4. WHICH YEARS SHOWED THE HIGHEST VOLATILITY?

Insight:

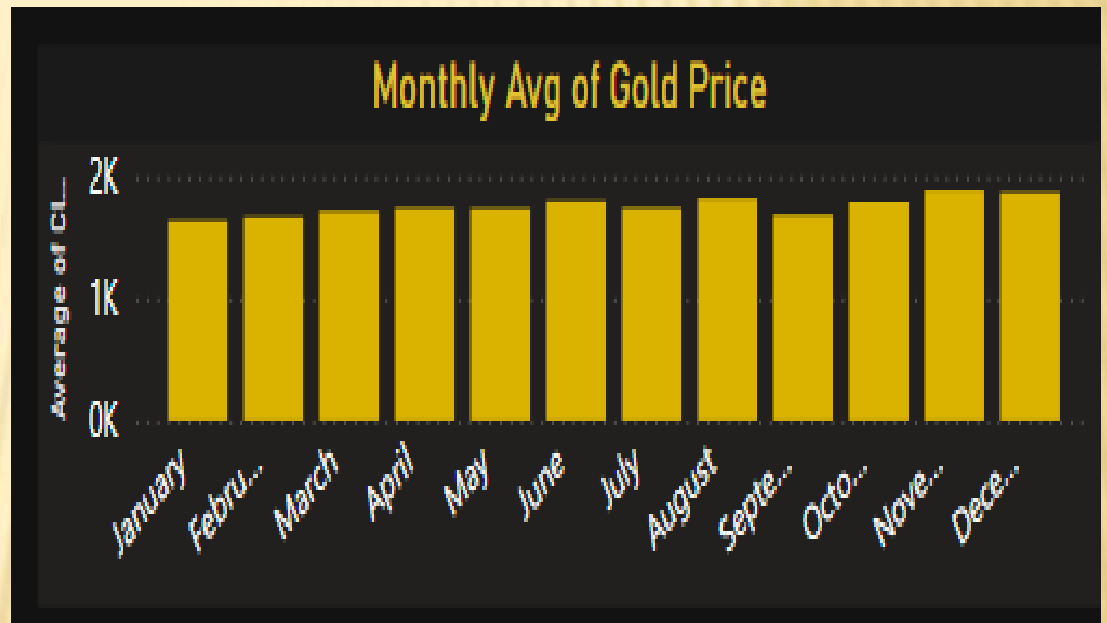
Volatility, measured using the weekly High-Low price range, was highest in years such as 2020 and 2025. These years were characterized by large fluctuations in price, indicating higher uncertainty and risk. High volatility reflects rapid changes in investor sentiment and external economic pressures. In contrast, earlier years exhibited lower volatility, suggesting more stable market conditions.



Q5. ARE THERE ANY SEASONAL PATTERNS IN GOLD PRICES?

Insight:

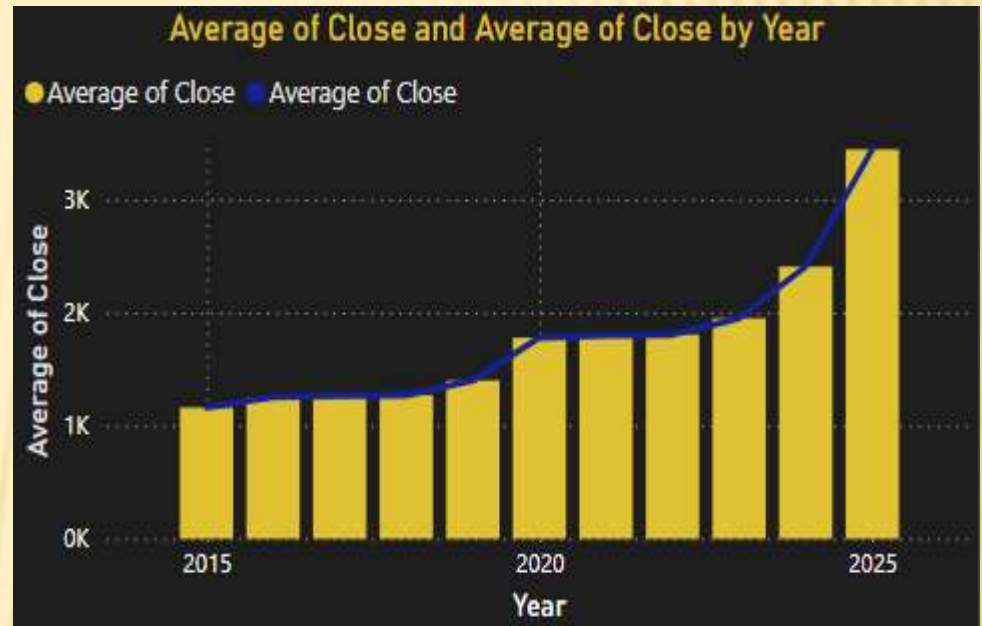
The analysis reveals that certain months consistently show higher average gold prices, indicating the presence of seasonal trends. These patterns may be influenced by factors such as festive demand, wedding seasons, and increased investment activity during specific times of the year. Such recurring patterns suggest that gold prices are not only influenced by global events but also by cyclical demand.



Q6. WHEN DID GOLD REACH NEW ALL-TIME HIGHS?

Insight:

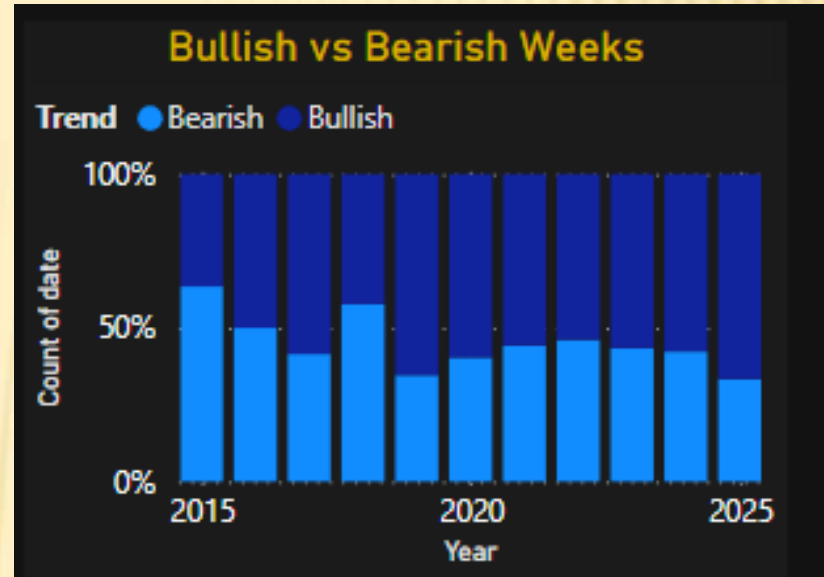
Gold prices breached new all-time highs primarily during and after 2020. These breakthroughs were clustered around periods of economic uncertainty, such as the pandemic and subsequent recovery phases. The increase in all-time highs indicates strong bullish momentum and reflects growing investor confidence in gold as a protective asset.



Q7. WHAT IS THE PROPORTION OF BULLISH VS BEARISH WEEKS?

Insight:

The analysis shows that a majority of the weeks are bullish, where the closing price is higher than the opening price. This indicates that upward price movements dominate the dataset. The presence of fewer bearish weeks suggests that selling pressure is relatively lower compared to buying interest, reinforcing the long-term upward trend of gold prices.



Q8. ARE THERE PATTERNS IN GAP OPENINGS?

Insight:

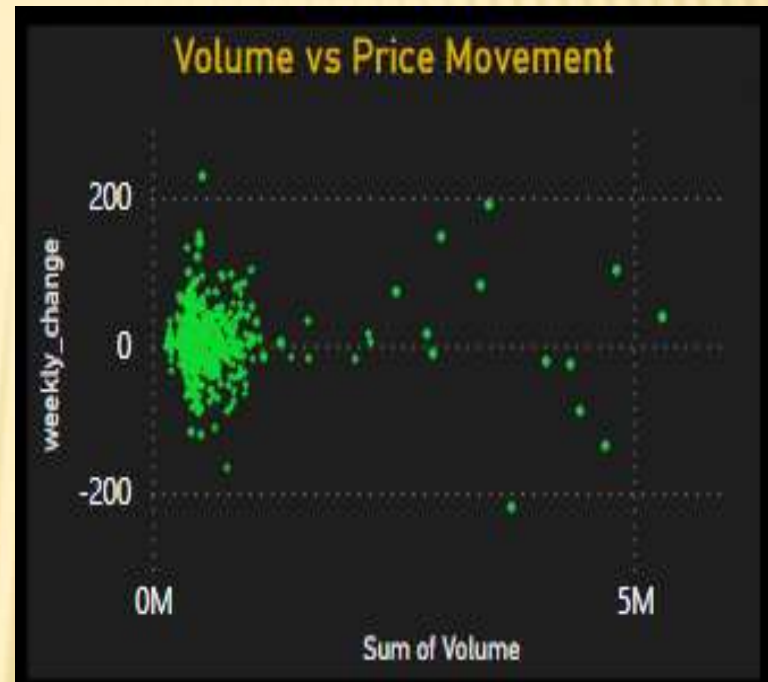
Gap openings, where the opening price differs significantly from the previous week's closing price, are observed during periods of high volatility. These gaps often occur due to sudden global news, economic announcements, or geopolitical events. The presence of such gaps highlights the responsiveness of gold prices to external shocks.

Year	Sum of Close	Sum of Open ▲
2015	60,297.81	60,456.40
2016	65,031.45	64,915.50
2018	65,971.48	66,023.30
2017	66,847.58	66,674.07
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2024	1,25,074.20	1,24,480.44
2025	1,65,173.13	1,64,008.40
Total	10,04,038.51	10,01,619.06

Q9. HOW DOES VOLUME RELATE TO PRICE MOVEMENT?

Insight:

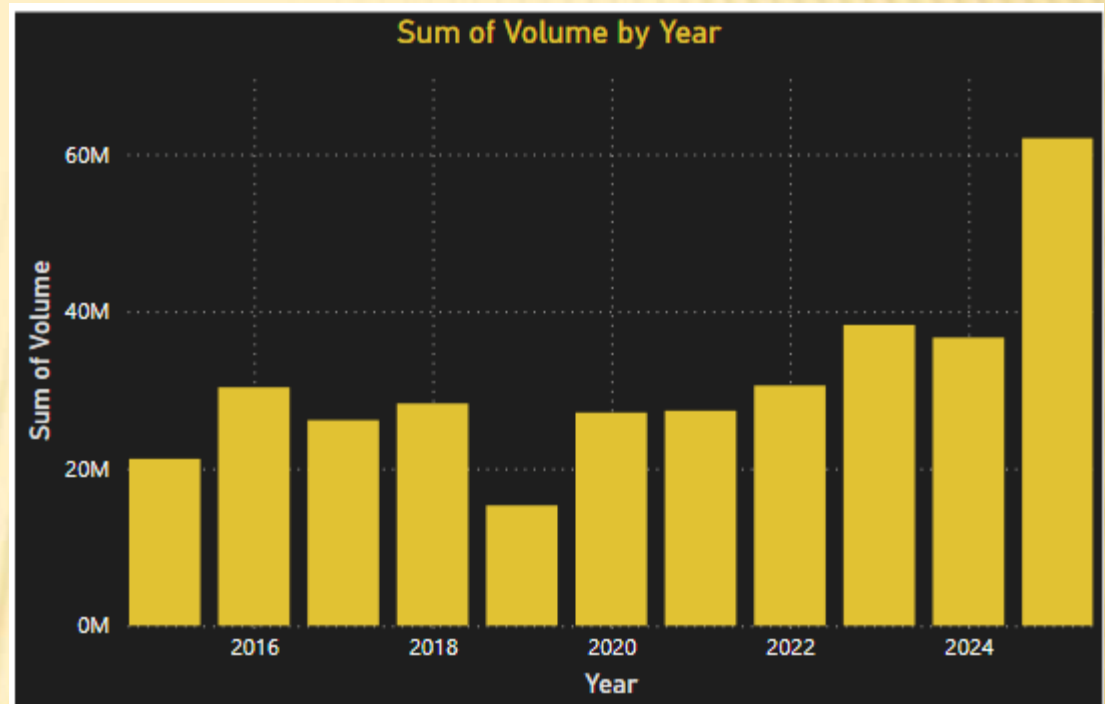
There is a noticeable relationship between trading volume and price movement. High-volume periods often coincide with large price swings, indicating strong market participation. When both price and volume increase together, it suggests a strong and confirmed trend. Conversely, low volume during price changes may indicate weak or temporary movements.



Q10. WHICH PERIODS HAD THE HIGHEST TRADING VOLUME?

Insight:

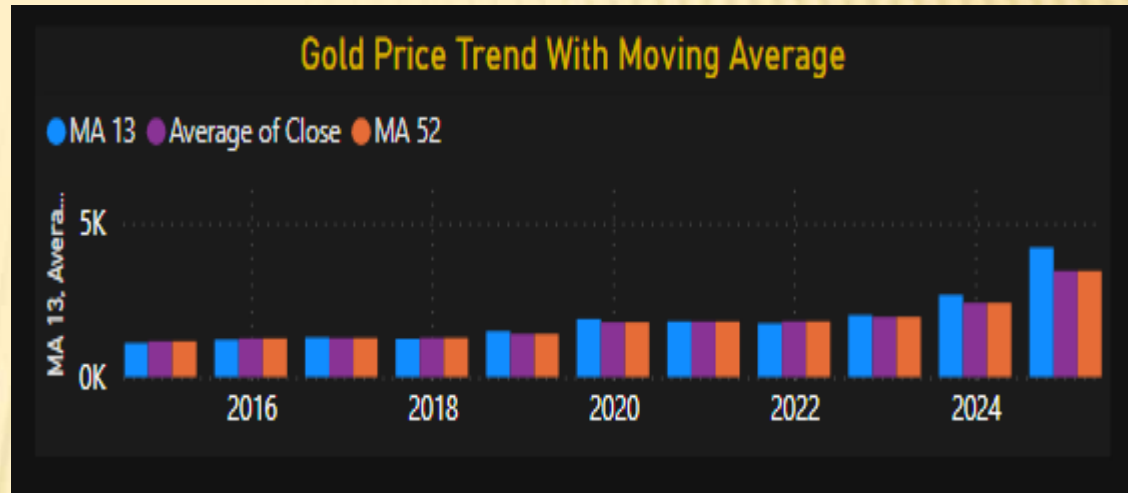
The highest trading volumes were recorded during periods of market uncertainty and strong price movements, particularly around 2020 and subsequent years. Increased volume reflects higher investor activity and interest in gold trading. These periods often align with major price rallies or corrections.



Q11. WHAT DO MOVING AVERAGES INDICATE?

Insight:

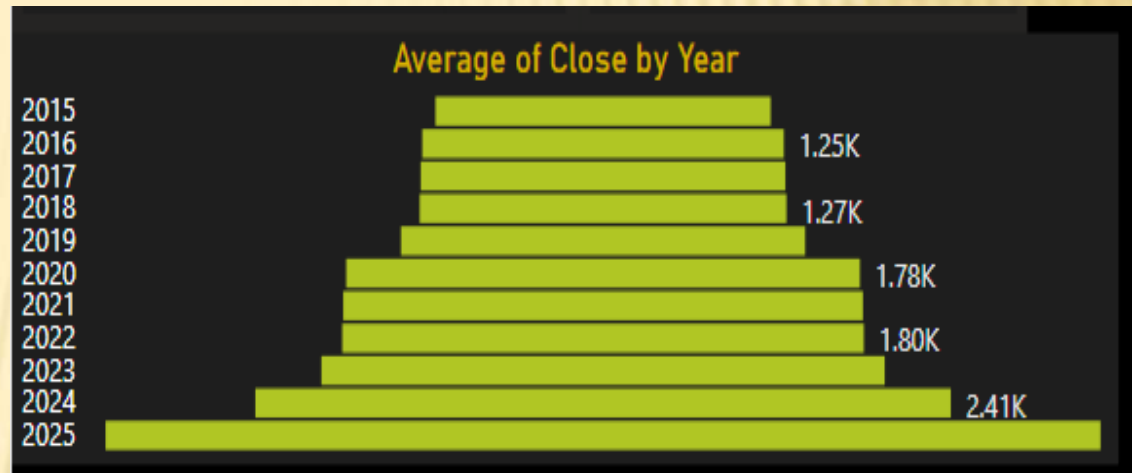
The 13-week and 52-week moving averages help identify short-term and long-term trends, respectively. When the price remains above these averages, it indicates a strong upward trend and acts as a support level. Conversely, when prices fall below the moving averages, it may signal a weakening trend or potential reversal.



Q13. HOW DID GOLD BEHAVE DURING MAJOR GLOBAL EVENTS?

Insight:

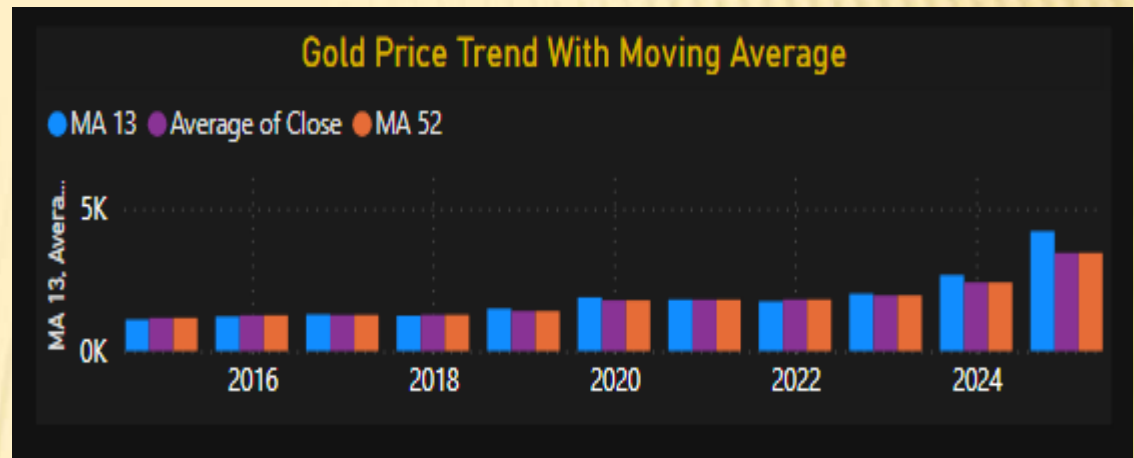
Gold prices reacted significantly to major global events. During the COVID-19 pandemic in 2020, prices surged due to increased demand for safe-haven assets. In 2022, inflation and interest rate changes caused heightened volatility. Geopolitical tensions in later years also contributed to price fluctuations. These observations confirm that gold is highly responsive to global economic conditions.



Q12. HOW OFTEN DID PRICE FALL BELOW THE 52-WEEK MOVING AVERAGE?

Insight:

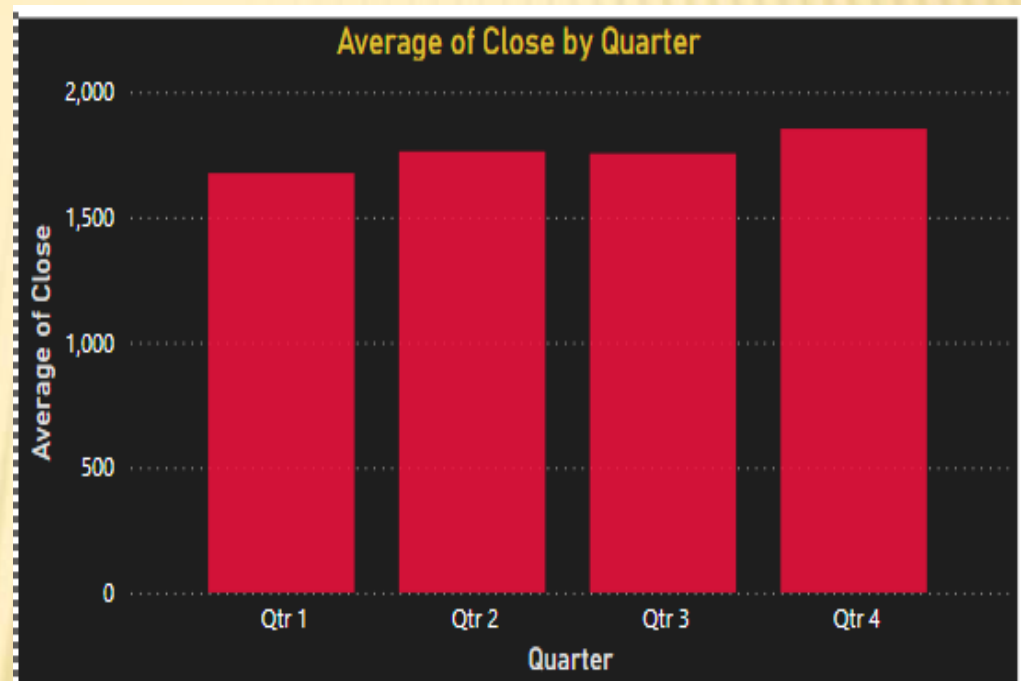
The price occasionally falls below the 52-week moving average, but such instances are usually temporary. In most cases, the price recovers quickly and moves back above the average, indicating strong long-term support. This behavior reinforces the resilience of gold prices over time.



Q14. WHICH QUARTERS PERFORM BEST?

Insight:

Certain quarters consistently show stronger performance compared to others. This may be due to seasonal demand, investment cycles, and global economic activities. Identifying strong and weak quarters helps in understanding periodic trends and planning investment strategies.



Q15. WHAT DO KEY KPIS REVEAL?

Insight:

Key performance indicators such as average closing price, maximum price, minimum price, and total volume provide a summary of gold market behavior. These KPIs highlight the overall growth, peak performance, and trading activity over the decade, offering a comprehensive overview of market performance.

1.76K	4.53K	1.06K	40.33	4.32K
avg_close	max_close	min_close	Volatility	Current Price

Q16. WHAT IS DRAWDOWN % IN GOLD PRICE ANALYSIS?

Insight:

The analysis of drawdown across the dataset shows that certain years, particularly those influenced by global economic instability, experienced higher drawdowns. These periods reflect increased market uncertainty and volatility. On the other hand, years with lower drawdowns indicate more stable price behavior and lower downside risk..

Drawdown percentage measures the decline in gold price from its highest point (peak) to a subsequent lowest point (trough) over a given period. It is an important risk indicator that helps in understanding the extent of losses an investor might face during market downturns.

